

Granularity Matters: Machine Learning and Sovereign Credit Ratings

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DEDICATION

To my incredible family: Granma, Mom, Sis, and Dad, thank you for letting me be, explore, and conquer the world. Your unwavering love and support have been my greatest strengths. This thesis is a testament to your faith in me. Thank you for everything.

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ABSTRACT

Sovereign credit ratings are crucial indicators of a country's creditworthiness, impacting borrowing costs and investor confidence. This thesis aims to enhance the prediction of sovereign credit ratings for the United States of America by employing advanced machine learning techniques. We integrate a diverse set of economic, political, social, and financial indicators into our models. Our methodology includes linear regression, ridge regression, ARMA, ARMAX, and decision trees, evaluated using RMSE and MAE metrics. The study reveals that incorporating governance and bond yield data significantly improves model accuracy. The best-performing models demonstrate a robust ability to predict credit ratings, suggesting that long-term and granular data provide substantial predictive power. These findings contribute to the literature by highlighting the importance of comprehensive data integration and advanced analytics in credit rating predictions.

Keywords: Credit Ratings, Machine Learning, Economic Indicators, Political Stability, Financial Analysis, United States

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Chapter 1

Introduction

1.1 The Significance of Credit Ratings

Sovereign credit ratings play a pivotal role in global financial markets by assessing the creditworthiness of countries. These ratings influence borrowing costs, investor confidence, and the overall economic stability of nations. Given their significant impact, accurately predicting sovereign credit ratings is essential for policymakers, investors, and financial analysts. Traditional econometric models have been widely used for this purpose; however, recent advancements in machine learning offer new opportunities to enhance prediction accuracy.

Credit ratings, assessed by prominent agencies such as Moody's, Standard & Poor's (S&P), and Fitch, are critical components of the global financial system. These ratings profoundly influence a country's access to international capital markets and determine the associated borrowing costs. High credit ratings enable governments to secure loans at lower interest rates, thereby reducing the financial burden of funding essential services and infrastructure projects. Conversely, low ratings increase borrowing costs and can lead to economic instability by restricting access to capital markets (Cantor & Packer, 1996).

Beyond sovereign debt, credit ratings affect investors' perceptions of risk across various financial markets, including stocks and bonds, influencing investment decisions (Reusens & Croux, 2017). Accurate credit ratings are essential for investors to manage risk effectively and allocate resources prudently. Additionally, credit ratings impact the borrowing costs of municipalities and corporations, thereby affecting

economic activities and growth at multiple levels (Alsakka, ap Gwilym, & Vu, 2014).

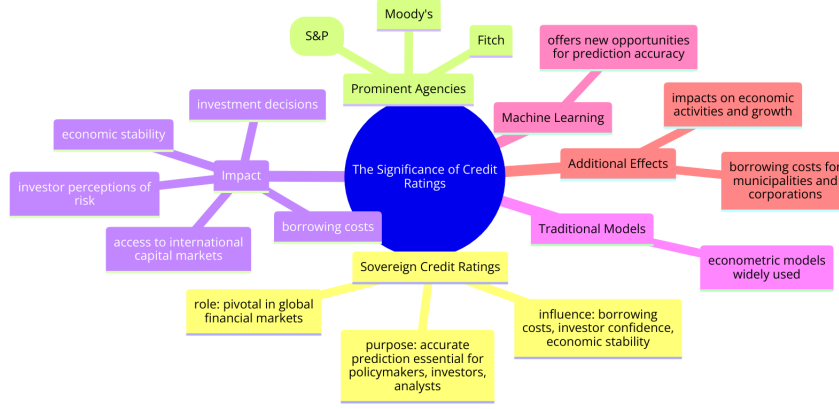


Figure 1.1: Significance of Credit Ratings

1.2 Motivation for the Study

The methodologies utilized by credit rating agencies have been subject to significant scrutiny, particularly in the aftermath of the 2008 financial crisis. Critics argue that these methodologies can be opaque, conflict-ridden, and occasionally detached from the underlying economic conditions of nations (White, 2010). This thesis is driven by the need for a transparent, responsive, and precise model that better integrates the economic, political, and social nuances influencing the credit ratings of the United States.

Advances in data science and machine learning present new opportunities to enhance the predictive accuracy of credit ratings. By leveraging extensive datasets and sophisticated algorithms, it is possible to uncover complex patterns and relationships that traditional methods might overlook (Wang, Xu, & Pusatli, 2015). This study aims to bridge the gap between traditional econometric approaches and modern data-driven techniques, providing a more robust framework for credit rating assessment.

1.3 Objectives of the Thesis

This thesis investigates the application of advanced machine learning techniques to predict the sovereign credit ratings of the USA. We utilize a comprehensive dataset encompassing economic indicators (such as GDP growth and inflation), bond yields, political stability, social factors, transparency, governance metrics, and banking sector health. By integrating these diverse data sources, we aim to capture the multifaceted nature of factors influencing credit ratings.

Our methodology involves constructing multiple models, including linear regression, ridge regression, ARMAX, and decision trees. We employ rigorous data preprocessing and normalization techniques to ensure data integrity. Each model's performance is evaluated using root mean square error (RMSE) and mean absolute error (MAE) metrics. Furthermore, we conduct a comparative analysis of models using annual and monthly data to understand the impact of data granularity on predictive accuracy.

The findings of this study indicate that models incorporating governance and bond yield data significantly outperform those relying solely on economic indicators (Kaufmann, Kraay, & Mastruzzi, 2010). This underscores the importance of political and institutional factors in credit rating assessments. Additionally, our results suggest that long-term and granular data enhance the models' predictive power, aligning with the hypothesis that sustained changes in socio-economic conditions are crucial for accurate credit rating predictions.

This research contributes to the existing literature by demonstrating the effectiveness of advanced machine learning techniques in credit rating predictions. It also provides practical insights for stakeholders in financial markets, emphasizing the need for comprehensive data integration and advanced analytics.

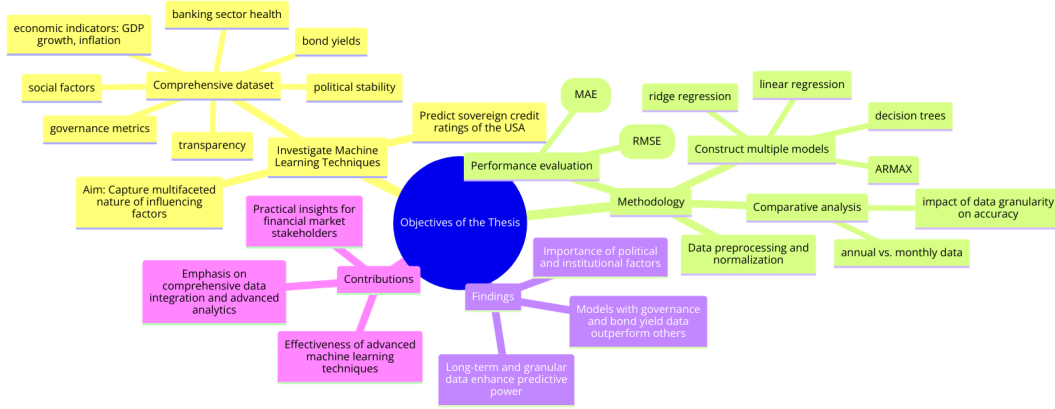


Figure 1.2: Thesis Objectives

1.4 Structure of the Thesis

The thesis is structured as follows:

- **Chapter 2: Literature Review** - Reviews the existing literature on credit rating methodologies and the application of machine learning models in finance.
- **Chapter 3: Datasets and Variables** - Describes the data sources, variable selection, and the rationale behind the chosen indicators.
- **Chapter 4: Methods** - Details the methodology employed in this study, including data collection, preparation, and the development of the predictive model.
- **Chapter 5: Annual vs Monthly Granularity: An Analysis of Credit Ratings** - Presents the results of the model testing and validation, comparing the performance of the new model with existing methodologies.
- **Chapter 6: Conclusions and Contributions** - Discusses the findings, their implications for credit rating practices and policy, and highlights the contributions of the thesis.

- **Chapter 7: Limitations and Further Recommendations** - Offers recommendations for future research and policy enhancements based on the findings, suggesting ways to further improve the accuracy and transparency of credit ratings.

Chapter 2

Literature Review

2.1 Credit Rating Models and Agency Methodologies

2.1.1 Introduction

Credit rating agencies (CRAs) play a pivotal role in the global financial system by providing assessments of the creditworthiness of entities and financial instruments. This review synthesizes extensive research on the methodologies employed by these agencies and the application of various statistical and machine learning models in finance, offering a thorough understanding from both technical and financial perspectives.

2.1.2 Statistical and Machine Learning Models in Finance

The financial industry extensively utilizes statistical models such as logistic regression and machine learning techniques, including decision trees and factor models, to assess credit risk and predict financial outcomes. Logistic regression, known for its efficacy in binary outcomes, is widely used in credit scoring to assess the probability of default (Takawira & Mwamba, 2022). Decision trees provide transparent, rule-based decision-making processes essential for modeling complex financial datasets, useful in market segmentation and risk assessment. Factor models, particularly useful in investment finance, analyze the influences on asset returns, aiding in portfolio construction and risk management (Connor & Korajczyk, 2009).

The integration of these models enhances predictive power and provides deeper

insights into financial behaviors. Hybrid models, such as those combining decision trees with logistic regression, have demonstrated improved performance in classification tasks within finance, effectively addressing linear relationships and interaction effects between variables (De Caigny, Coussement, & De Bock, 2018; M. Kim, 2009).

2.1.3 Credit Rating Agency Methodologies

Credit ratings result from complex analyses involving qualitative and quantitative assessments of an issuer's financial health and broader economic indicators. Agencies employ sophisticated econometric models that integrate financial ratios, market trends, and macroeconomic data to derive credit scores. These methodologies aim to provide stability in ratings through economic cycles, focusing on long-term creditworthiness rather than short-term market fluctuations (Fitch Ratings, 2023; Standard & Poor's Financial Services LLC, 2019).

2.1.4 Socio-Economic and Institutional Factors

In addition to financial metrics, socio-economic and political conditions significantly influence credit ratings. Factors such as political stability, governance quality, regulatory frameworks, and economic policies collectively impact an issuer's credit risk. CRAs adjust their methodologies to reflect potential impacts from these macroeconomic and socio-political dynamics, which can mitigate or amplify the perceived risk associated with an issuer (Kozmenko & Plastun, 2012).

2.1.5 Temporal Interpolation and Aggregation

Temporal aggregation and interpolation are crucial techniques in financial studies for handling time series data, particularly when dealing with missing data or creating aggregated time series from high-frequency data.

Temporal Aggregation in Financial Studies: Silvestrini and Veredas (2008) provide a comprehensive survey on methods of temporal aggregation and their implications for time series modeling in econometrics and finance. The study discusses how different aggregation methods affect the properties and performance of univariate and multivariate time series models (Silvestrini & Veredas, 2008).

Time Series Interpolation: Interpolation methods are used to estimate missing values within a time series. A review article (Lepot, Aubin, & Clemens, 2017) provides an overview of existing interpolation methods, their performance criteria, and uncertainty assessment, offering valuable insights for selecting appropriate techniques for specific financial datasets.

2.1.6 Data Granularity and Predictive Accuracy

The granularity of data plays a crucial role in the accuracy of predictive models. Several studies have explored the impact of data frequency on forecasting performance. For instance, (Brogaard, Hendershott, & Riordan, 2014) examined how high-frequency trading data affects market quality and predictive accuracy, highlighting the benefits of granular data in financial modeling. Similarly, (Faust & Wright, 2013) demonstrated the significance of data frequency and timeliness in improving inflation forecasts, which can be extended to credit rating predictions. Moreover, (Ghysels, Sinko, & Valkanov, 2007; Clements & Galvão, 2008) discussed the advantages of using mixed frequency data in macroeconomic forecasting, suggesting that incorporating both annual and monthly data can enhance model performance.

2.1.7 Identified Gaps and Contributions

Despite extensive research on sovereign credit rating predictions, several critical gaps remain. This thesis addresses these gaps through the following contributions.

Integration of Diverse Indicators

Existing studies predominantly focus on economic indicators, often neglecting the multifaceted nature of factors influencing sovereign credit ratings. This thesis fills this gap by incorporating a wide range of indicators, including economic, political, social, governance, and financial sector health metrics, providing a more nuanced understanding of the determinants of sovereign credit ratings (Cantor & Packer, 1996; Mellios & Paget-Blanc, 2006).

Advanced Machine Learning Techniques

Traditional econometric models have been extensively used in the literature, yet there is limited application of advanced machine learning techniques. This thesis bridges this gap by employing models such as ARMAX, decision trees, and ensemble methods, showcasing their potential to enhance predictive accuracy (Archer, Biglaiser, & DeRouen, 2007; Afonso, Gomes, & Rother, 2011).

Comparative Analysis of Model Performance

Few studies provide a detailed comparative analysis of various predictive models. This thesis conducts a rigorous comparison of different models based on RMSE and MAE metrics, highlighting the strengths and weaknesses of each approach and providing clear guidance on model selection for future research (Hill, Brooks, & Faff, 2010).

Focus on Major Economies

While many studies encompass a broad range of countries or focus on emerging markets, there is a lack of focused research on major economies like the USA, France, and Germany. This thesis specifically examines these key economies, offering detailed insights into their credit rating dynamics and contributing valuable country-specific knowledge (Alsakka et al., 2014).

Impact on Governance and Institutional Quality

Although the role of governance and institutional quality is recognized, it is often not quantitatively integrated into predictive models. This thesis addresses this by incorporating governance indicators into the models, demonstrating their significant impact on credit rating predictions and providing a more comprehensive understanding of the factors at play (S.-J. Kim & Wu, 2008; Gärtner, Griesbach, & Jung, 2011).

By addressing these gaps, this thesis not only advances the academic understanding of sovereign credit rating predictions but also offers practical insights for policymakers and financial analysts. The findings underscore the importance of a multifaceted and data-driven approach to credit rating analysis.

2.2 Machine Learning Models

2.2.1 Introduction

This section explores various regression techniques and time series models, focusing on linear regression, ridge regression, autoregressive moving average (ARMA), autoregressive moving average with exogenous inputs (ARMAX), and decision tree methods. These methods are foundational in data science for predicting outcomes and uncovering relationships within data. By examining their historical development, mathematical underpinnings, applications, and limitations, we aim to provide a comprehensive understanding of their roles in data science.

2.2.2 Linear Regression

Historical Background: Linear regression is one of the oldest and most widely used statistical techniques, dating back to the early 19th century with contributions from Carl Friedrich Gauss and Adrien-Marie Legendre (Stigler, 1986).

Key Concepts and Mathematical Formulation: Linear regression models the relationship between a dependent variable Y and one or more independent variables X using the equation $Y = \beta_0 + \beta_1 X_1 + \dots + \beta_n X_n + \epsilon$, where β represents the coefficients and ϵ is the error term (“Linear Models for Regression”, 2006).

Applications: Linear regression is extensively used in economics, biology, engineering, and social sciences for predictive analysis and inferential statistics (Yan, 2009).

Limitations and Advancements: While simple and interpretable, linear regression assumes a linear relationship, which may not always hold. Advancements include the development of techniques to handle multicollinearity and heteroscedasticity (Tibshirani, 1996).

2.2.3 Ridge Regression

Introduction: Ridge regression, developed to address some limitations of linear regression, especially multicollinearity, introduces a regularization parameter to shrink coefficients (Tibshirani, 1996).

Comparison with Linear Regression: Unlike linear regression, ridge regression adds a penalty equivalent to the square of the magnitude of coefficients to the loss function, effectively reducing variance at the cost of introducing bias (Hoerl & Kennard, 1970).

Mathematical Formulation: Ridge regression modifies the ordinary least squares (OLS) loss function to include a regularization term:

$$\text{Minimize } \|Y - X\beta\|^2 + \lambda\|\beta\|^2$$

where λ is the regularization parameter.

Applications and Case Studies: Common in situations where predictors are

highly correlated, such as genomics and finance. Case studies demonstrate its effectiveness in improving predictive performance (Hoerl & Kennard, 1970; Yan, 2009).

Advancements and Limitations: Extensions like the LASSO and Elastic Net address the need for variable selection and model complexity management (Tibshirani, 1996).

2.2.4 Autoregressive Moving Average (ARMA)

Overview of Time Series Analysis: Time series models are essential for analyzing data points collected or recorded at specific time intervals (Box, Jenkins, Reinsel, & Ljung, 2015).

Introduction to ARMA Models: ARMA models combine autoregressive (AR) and moving average (MA) components to capture various aspects of temporal correlations (Rao, Rao, & Rao, 2012).

Mathematical Formulation: An ARMA model is represented as

$$Y_t = c + \sum_{i=1}^p \phi_i Y_{t-i} + \sum_{j=1}^q \theta_j \epsilon_{t-j} + \epsilon_t$$

where p and q are the orders of the AR and MA components, respectively.

Applications in Forecasting: Widely used in economic forecasting, signal processing, and environmental modeling due to its ability to model complex time-dependent structures (Box et al., 2015).

Strengths, Limitations, and Improvements: ARMA models are powerful but require stationarity of the time series. Extensions like ARIMA and SARIMA address non-stationarity and seasonal effects (Box et al., 2015; Rao et al., 2012).

2.2.5 Autoregressive Moving Average with Exogenous Inputs (ARMAX)

Introduction to ARMAX Models: The ARMAX model extends the ARMA model by incorporating exogenous inputs, providing a more comprehensive framework for time series analysis when external variables are believed to influence the dependent variable (Chan, 2011).

Mathematical Formulation: An ARMAX model is represented as

$$Y_t = c + \sum_{i=1}^p \phi_i Y_{t-i} + \sum_{j=1}^q \theta_j \epsilon_{t-j} + \sum_{k=1}^r \beta_k X_{t-k} + \epsilon_t$$

where p , q , and r are the orders of the AR, MA, and exogenous input components, respectively.

Applications in Finance: ARMAX models are particularly useful in financial econometrics where external variables, such as interest rates, stock indices, or economic indicators, are expected to have significant impacts on the time series being modeled (Enders, 2014).

Strengths and Limitations: The inclusion of exogenous variables allows ARMAX models to capture more complex dynamics and improve forecast accuracy. However, selecting appropriate exogenous variables and determining their lag structures can be challenging and requires domain knowledge and empirical testing (Chan, 2011).

Empirical Studies: Numerous studies have demonstrated the effectiveness of ARMAX models in financial time series forecasting (Noverianto, Suharsono, & Prastyo, 2023; Lapitskaya, Eratalay, & Sharma, 2022).

2.2.6 Decision Tree

Introduction to Decision Trees: Decision trees are non-parametric models used for both classification and regression tasks, providing a clear visualization of decision-making processes (Breiman, Friedman, Olshen, & Stone, 2017).

Historical Development and Key Contributors: The development of decision trees can be traced back to the 1960s, with significant contributions from researchers like Ross Quinlan and Leo Breiman.

Algorithm and Mathematical Basis: Decision trees split data based on feature values, optimizing splits using metrics like Gini impurity or information gain (Breiman et al., 2017).

Applications in Classification and Regression: Extensively used in various domains, from medical diagnosis to financial analysis, due to their interpretability and ease of use (Breiman et al., 2017).

Advantages and Limitations: Decision trees are easy to understand and implement but can suffer from overfitting and instability. Techniques like pruning, bagging, and boosting address these issues (Breiman et al., 2017).

2.2.7 Comparative Analysis

Each of these methods offers unique strengths and is suited to different types of problems. Linear and ridge regressions are best for problems with a clear linear relationship, while ARMA and ARMAX models excel in time series forecasting, with ARMAX providing enhanced capabilities by incorporating exogenous variables. Decision trees provide a versatile and interpretable model for both classification and regression tasks but require careful tuning to avoid overfitting.

2.2.8 Conclusion

This literature review highlights the importance of understanding various regression and time series methods, their applications, and limitations. Selecting the appropriate model depends on the specific characteristics of the dataset and the problem at hand, underscoring the need for a deep understanding of these foundational techniques in data science.

Chapter 3

Datasets and Variables

3.1 Datasets

This section provides an overview of the datasets utilized in this study, including their sources, key features, and applications. Each dataset plays a crucial role in analyzing different aspects of economic and financial indicators.

3.1.1 World Development Indicators (2023)

Source: World Bank (World Bank, 2023)

Dataset: World Development Indicators

URL: <https://data.worldbank.org/indicator>

Description:

The World Development Indicators (WDI) dataset is a comprehensive collection of global development data compiled by the World Bank. It aggregates information from officially recognized international sources, providing national, regional, and global estimates. This dataset supports a wide range of economic, social, and environmental analyses, facilitating informed decision-making and policy formulation.

Key Features:

- **Coverage:** Encompasses over 200 economies worldwide, offering a global perspective.
- **Time Frame:** Data is collected annually and updated regularly to reflect the most recent information.

- **Variables:** Contains over 1,400 indicators, covering diverse aspects such as economic performance, health, education, environment, and infrastructure.
- **Applications:** Utilized extensively in economic analysis, policy-making, academic research, and educational purposes.

Examples of Indicators:

- GDP per capita
- Life expectancy at birth
- Adult literacy rate
- Access to electricity (% of population)
- CO₂ emissions (metric tons per capita)

3.1.2 WRDS U.S. Treasury Inflation Data (2024)

Source: Wharton Research Data Services (WRDS) (Wharton Research Data Services, 2024)

Dataset: WRDS U.S. Treasury Inflation Data

URL: <https://wrds.wharton.upenn.edu>

Description:

The WRDS U.S. Treasury Inflation Data offers an extensive collection of U.S. inflation metrics and economic indicators sourced from the U.S. Treasury. This dataset is critical for examining inflationary trends and assessing monetary policy impacts, providing detailed insights necessary for both academic and professional financial research.

Key Features:

- **Coverage:** Focuses exclusively on the United States.
- **Time Frame:** Includes both monthly and annual data, ensuring up-to-date and historical perspectives.
- **Variables:** Features essential economic indicators such as the Consumer Price Index (CPI), various inflation rates, Treasury yields, and break-even inflation rates.
- **Applications:** Essential for financial research, economic analysis, policy evaluation, and academic studies concerning inflation and economic performance.

Examples of Indicators:

- CPI for all urban consumers (CPI-U)
- Treasury Inflation-Protected Securities (TIPS) yields
- Break-even inflation rates
- Nominal and real interest rates
- Inflation expectations

3.1.3 United States Credit Rating

Source: World Government Bonds

URL: <https://www.worldgovernmentbonds.com/credit-rating/united-states/>

Description:

The United States Credit Rating dataset provides up-to-date credit ratings issued by major credit rating agencies. This dataset is vital for understanding the credit-worthiness of the U.S. government, influencing investment decisions, and evaluating economic stability.

Key Features:

- **Coverage:** Provides ratings from leading credit rating agencies such as Standard & Poor's, Moody's, and Fitch.
- **Time Frame:** Updated regularly to reflect the most current assessments.
- **Variables:** Includes long-term and short-term credit ratings, rating outlooks, and historical rating changes.
- **Applications:** Crucial for investment analysis, risk assessment, economic research, and policy development.

Examples of Indicators:

- Long-term credit rating (e.g., AA+ by Standard & Poor's)
- Short-term credit rating (e.g., A-1+ by Standard & Poor's)
- Rating outlook (e.g., stable, positive, negative)

3.2 Variable Selection

3.2.1 The Rationale

The assessment of a country's credit rating by agencies like Fitch and Standard & Poor's (S&P) involves evaluating various economic and political indicators. Here are some of the pivotal factors:

1. **Economic Indicators:** Economic performance indicators such as GDP per capita, GDP growth rate, inflation rates, external debt, and overall economic development are significant determinants of a country's credit rating. These factors reflect the economic health and the potential for growth, influencing the

perceived risk associated with a country's ability to meet its financial obligations. Studies have shown that robust economic growth and low inflation are positively correlated with higher credit ratings (Cantor & Packer, 1996; Erdem & Varlı, 2014).

2. **Government Debt to GDP Ratio:** This ratio is a critical measure of a country's fiscal health. A lower debt-to-GDP ratio generally indicates a better capacity of the government to meet its debt obligations without defaulting, leading to a higher credit rating (AlKulaib & AlAli, 2021).
3. **Political and Institutional Factors:** The stability and effectiveness of political institutions significantly impact credit ratings. Countries with stable governments and efficient, transparent institutions tend to receive higher credit ratings, as these factors are crucial for economic stability and governance (Archer et al., 2007).
4. **Banking Sector Health:** The health of a country's banking system also plays a vital role in its credit rating. A sound, well-regulated, and stable banking system reduces financial risk and contributes positively to a country's rating. The resilience of banks during economic downturns and their ability to manage non-performing loans are particularly important indicators (Cuadros-Solas & Salvador Muñoz, 2021).
5. **Default History:** Countries with a history of defaulting on their debt are viewed as riskier investments. Credit rating agencies use historical data to assess the likelihood of future defaults. Repeated defaults tend to lower credit ratings significantly (Cantor & Packer, 1996).
6. **Transparency and Corruption:** The level of transparency in financial and

governmental institutions and the perceived corruption within a country can greatly influence its credit ratings. Higher transparency and lower corruption levels generally improve a country's credit rating (Mellios & Paget-Blanc, 2006).

Variable	Description
Gross savings (current US\$)	Total savings generated within the country.
Trade (% of GDP)	The sum of exports and imports of goods and services.
GDP growth (annual %)	Annual percentage growth rate of GDP.
GDP (current US\$)	Gross domestic product at current market prices.
Net lending (+) / net borrowing (-) (% of GDP)	Net financial position of the government.
Foreign direct investment, net inflows (% of GDP)	Investment from foreign entities in the domestic economy.
Current account balance (% of GDP)	Difference between exports and imports of goods and services.
Inflation, consumer prices (annual %)	Annual percentage change in the cost to the average consumer of acquiring a basket of goods and services.
Unemployment, total (% of total labor force) (national estimate)	Percentage of the labor force that is unemployed.

Table 3.1: Economic Variables considered from the WDI dataset

Variable	Description
Bank liquid reserves to bank assets ratio (%)	Proportion of a bank's assets held in liquid form.
Bank capital to assets ratio (%)	Ratio of a bank's capital to its total assets.
Claims on central government, etc. (% GDP)	Financial claims on the central government as a percentage of GDP.
Total reserves (includes gold, current US\$)	Total reserves held by the country, including gold.
Bank nonperforming loans to total gross loans (%)	Proportion of nonperforming loans in a bank's total gross loan portfolio.

Table 3.2: Banking Sector Health variables considered from the WDI dataset

Indicator	Description
Control of Corruption: Estimate	Measures the extent to which public power is exercised for private gain.
Government Effectiveness: Estimate	Quality of public services, civil service, and policy formulation.
Political Stability and Absence of Violence/Terrorism: Estimate	Likelihood of political instability and violence.
Regulatory Quality: Estimate	Ability of the government to formulate and implement sound policies.
Rule of Law: Estimate	Extent to which agents have confidence in and abide by the rules of society.
Voice and Accountability: Estimate	Extent to which citizens can participate in selecting their government.

Table 3.3: Governance Indicators considered from the WDI dataset

Variable	Description
b30ret, b30ind	Return and index values for 30-year U.S. Treasury bonds.
b20ret, b20ind	Return and index values for 20-year U.S. Treasury bonds.
b10ret, b10ind	Return and index values for 10-year U.S. Treasury bonds.
b7ret, b7ind	Return and index values for 7-year U.S. Treasury bonds.
b5ret, b5ind	Return and index values for 5-year U.S. Treasury bonds.
b2ret, b2ind	Return and index values for 2-year U.S. Treasury bonds.
b1ret, b1ind	Return and index values for 1-year U.S. Treasury bonds.
cpiret, cpiind	Return and index values for the Consumer Price Index (CPI).

Table 3.4: WRDS U.S. Treasury Inflation Data Variables considered

Chapter 4

Methods

4.1 Credit Rating Agencies - Methods

Fitch Ratings	S&P Global Ratings
Quantitative Model	
Multiple regression model using 18 key variables grouped into four pillars:	Combines quantitative models with qualitative assessments; rates five key factors on a six-point scale:
• Structural Features • Macroeconomic Performance, Policies, and Prospects • Public Finances • External Finances	• Institutional Assessment • Economic Assessment • External Assessment • Fiscal Assessment • Monetary Assessment
Qualitative Overlay (QO)	
Adjusts the SRM output to account for factors not fully captured by the model; adjustments up to ± 2 notches per pillar, with an overall range of ± 3 notches.	Adjusts based on qualitative factors and expert judgment; final rating can be adjusted by up to ± 3 notches based on supplemental factors.
Key Variables and Weights	
Structural Features (53.5%)	Institutional Assessment
Macroeconomic Performance (10.1%)	Economic Assessment
Public Finances (18.5%)	External Assessment
External Finances (17.9%)	Fiscal Assessment
	Monetary Assessment

Table 4.1: Summary of Fitch’s and S&P’s Sovereign Rating Methodologies

Fitch Ratings

Quantitative Model

Uses a multiple regression model (SRM) with 18 key variables across four pillars.

Mathematical Framework

$$\text{SRM Score} = \sum (\text{Coefficient}_i \times \text{Variable}_i)$$

Adjusted by qualitative overlay (QO):

$$\text{Final Rating} = \text{SRM Score} + \sum (\text{Qualitative Adjustments})$$

S&P Global Ratings

Quantitative Model

Combines quantitative models with qualitative assessments; rates five key factors on a six-point scale.

Mathematical Framework

$$\text{Rating Factor Score} = \sum (\text{Coefficient}_i \times \text{Variable}_i)$$

Adjusted by qualitative factors and expert judgment:

$$\text{Final Rating} = \text{Indicative Rating} + \sum (\text{Qualitative Adjustments})$$

Table 4.2: Condensed Mathematical Frameworks of Fitch’s and S&P’s Sovereign Rating Methodologies

Credit rating agencies play a crucial role in assessing the creditworthiness of sovereign countries by evaluating their ability to meet financial obligations. They employ a variety of methodologies that combine both quantitative models and qualitative assessments to determine sovereign credit ratings (Standard & Poor’s Financial Services LLC, 2019; Fitch Ratings, 2023).

4.2 Methods Used in the Thesis

In this study, we aim to enhance the prediction of sovereign credit ratings for the United States by leveraging advanced machine learning techniques. Given the constraints of our dataset, which includes annual data with approximately 40-50 observations and interpolated monthly data with around 400 observations, we carefully selected models that are well-suited for low data regimes. The models chosen for this study are linear regression, ridge regression, ARMAX, and decision trees. Each model offers unique advantages in handling the characteristics of our dataset.

4.2.1 Linear Regression

Linear regression is one of the simplest and most interpretable models, making it an excellent baseline for our analysis. It provides clear insights into the relationship between predictors and the target variable, which is valuable in financial analysis where understanding the impact of individual predictors is crucial. Given the limited number of annual observations (40-50), linear regression's low data requirement is particularly advantageous (Wooldridge, 2003).

4.2.2 Ridge Regression

Ridge regression addresses the issue of multicollinearity, which is common in financial datasets where predictors can be highly correlated. By including a regularization term, ridge regression penalizes large coefficients, leading to more stable and generalizable models. This is especially important in our context of limited data, helping to prevent overfitting (Hoerl & Kennard, 1970).

4.2.3 ARMAX (Autoregressive Moving Average with Exogenous Variables)

The ARMAX model is well-suited for capturing temporal dependencies in time series data. By incorporating both autoregressive and moving average components, along with exogenous variables, ARMAX can model complex relationships more effectively. This is particularly beneficial for our monthly data, which includes around 400 observations and captures more granular temporal patterns (Box et al., 2015).

4.2.4 Decision Trees

Decision trees are capable of capturing non-linear relationships between predictors and the target variable. This makes them particularly useful in financial contexts where such non-linearities are common. Additionally, decision trees provide a visual representation of the decision-making process and highlight the most important features influencing the predictions. They are also adept at handling missing data and outliers, which can be beneficial when dealing with interpolated monthly data (Breiman et al., 2017).

4.2.5 Data Collection and Preparation

The analysis utilized four primary data sources:

- **World Development Indicators, 2023:** Data sourced from the World Bank, containing various economic indicators, government debt to GDP ratio, political and institutional factors, banking sector health, and transparency and corruption relevant to the United States from 1980 onwards.
- **WRDS U.S. Treasury Inflation Data (2024):** Monthly US Treasury Inflation-Protected Securities (TIPS) data obtained from WRDS.

- **Credit Rating Data:** Fitch and S&P credit rating data.

Loading Datasets

The datasets were loaded into the analysis environment using Pandas. This involved reading the CSV files for economic data and bond data and creating a structured format for credit ratings data.

Credit Ratings Data Preparation

Step	Description
Data Frame Creation	The annual credit ratings data was initially stored in a dictionary format and then converted into a Pandas DataFrame for easier manipulation and analysis.
Year Range Generation	A DataFrame containing a range of years from the specified start year to 2022 was created to ensure the dataset covers a consistent timeline.
Merging and Forward-Filling	The year range DataFrame was merged with the credit ratings DataFrame, and any missing values were forward-filled to maintain data continuity.
Mapping Ratings to Numerical Values	Credit ratings from different agencies were mapped to numerical values using a predefined scheme to facilitate quantitative analysis.
Index Setting and Resampling	The DataFrame index was set to a date range from the start year to 2022 with annual frequency, and then resampled to monthly frequency using spline interpolation to fill in the gaps.

Table 4.3: Steps in Credit Ratings Data Preparation

Credit Rating (Positive outlook)	Value	Credit Rating (Negative outlook)	Value
AAA(positive)	0.1	AAA(negative)	0.2
AA+(positive)	0.3	AA+(negative)	0.4
AA(positive)	0.5	AA(negative)	0.6
AA-(positive)	0.7	AA-(negative)	0.8
A+(positive)	0.9	A+(negative)	1.0
A(positive)	1.1	A(negative)	1.2
A-(positive)	1.3	A-(negative)	1.4
BBB+(positive)	1.5	BBB+(negative)	1.6
BBB(positive)	1.7	BBB(negative)	1.8
BBB-(positive)	1.9	BBB-(negative)	2.0
BB+(positive)	2.1	BB+(negative)	2.2
BB(positive)	2.3	BB(negative)	2.4
BB-(positive)	2.5	BB-(negative)	2.6

Table 4.4: Credit Rating Mapping

The final output includes:

- **Annual Credit Ratings DataFrame:** Contains annual credit ratings with numerical values.
- **Monthly Credit Ratings DataFrame:** Resampled and interpolated monthly ratings from annual data.

Economic Data Preparation

Step	Description
Filtering for the United States	The dataset containing global economic indicators was filtered to retain only the data relevant to the United States.
Selection of Relevant Indicators	A subset of economic indicators pertinent to credit scoring was selected, including GDP per capita, inflation, FDI, unemployment, and others. Additional banking sector health indicators were included if the start year was 2001 or later.
Cleaning and Transposing Data	The dataset was cleaned by removing irrelevant columns and transposed to have indicators as columns and years as rows.
Selecting Data Range	The dataset was filtered to include data from the specified start year to 2022.
Index Setting and Resampling	The DataFrame index was set to a date range from the start year to 2022 with annual frequency, and then resampled to monthly frequency using spline interpolation to fill in the gaps.

Table 4.5: Steps in Economic Data Preparation

The final output includes:

- **Annual Economic Data DataFrame:** Contains selected economic indicators with annual frequency.
- **Monthly Economic Data DataFrame:** Resampled and interpolated monthly values from annual data.

Governance Indicators Data Preparation

Step	Description
Filtering for the United States	The global governance indicators dataset was filtered to retain only records relevant to the United States.
Selection of Relevant Indicators	A subset of governance indicators was selected, including control of corruption, government effectiveness, political stability, regulatory quality, rule of law, and voice and accountability.
Cleaning and Transposing Data	The dataset was cleaned by removing irrelevant columns and then transposed to have indicators as columns and years as rows.
Selecting Data Range	The dataset was filtered to include records from the specified start year to 2022.
Index Setting and Resampling	The DataFrame index was set to a date range from the start year to 2022 with annual frequency, and then resampled to monthly frequency using spline interpolation.
Handling Missing Values	Missing values in the annual governance indicators data were forward-filled to ensure completeness.

Table 4.6: Steps in Governance Indicators Data Preparation

The final output includes:

- **Annual Governance Indicators DataFrame:** Contains selected governance indicators with annual frequency.
- **Monthly Governance Indicators DataFrame:** Resampled and interpolated monthly values from annual data.

Treasury Bond Data Preparation

Step		Description
Adding Column	Year	A new column was created to extract the year from the date column for filtering purposes.
Filtering by Year	Data	The dataset was filtered to include only records from the specified start year up to 2023.
Setting Index		The index of the DataFrame was set to the date column to facilitate time series analysis.
Removing Irrelevant Columns		Irrelevant columns, such as the newly added 'year' column, were removed to clean the dataset.
Resampling Data		The dataset was resampled to a monthly frequency, calculating the mean values for each month to standardize the data.

Table 4.7: Steps in Treasury Bond Data Preparation

The final output includes:

- **Annual Treasury Bond DataFrame:** Contains resampled treasury bond data with monthly frequency.
- **Monthly Treasury Bond DataFrame:** Original filtered monthly data with the specified date range.

4.2.6 Merging and Normalizing Data

Data Merging and Normalization

The process of merging and normalizing the datasets integrates various data sources and standardizes the scales of the variables.

Step		Description
Dropping Columns	Year	Removing the 'Year' columns from the economic or bond datasets to prevent duplication during the merging process.
Merging DataFrames		Merging the credit ratings data with economic data, bond data, and a combination of both to create three merged DataFrames: <code>econ_credit</code> , <code>bond_credit</code> , and <code>econ_bond_credit</code> .
Normalization		Normalizing each merged DataFrame using <code>StandardScaler</code> to standardize the variables, ensuring they have a mean of 0 and a standard deviation of 1.

Table 4.8: Steps in Basic Merging and Normalization

The final output includes three normalized DataFrames:

- `df1_scaled`: Normalized credit ratings and economic data.
- `df2_scaled`: Normalized credit ratings and bond data.
- `df3_scaled`: Normalized combined dataset of credit ratings, economic data, and bond data.

Extended Merging and Normalization

Step		Description
Dropping Columns	Year	Removing the 'Year' columns from the economic, governance (WGI), or bond datasets to prevent duplication during the merging process.
Merging DataFrames		Merging the credit ratings data with various combinations of economic, governance (WGI), and bond data to create seven merged DataFrames: <code>econ_credit</code> , <code>wgi_credit</code> , <code>bond_credit</code> , <code>econ_wgi_credit</code> , <code>econ_bond_credit</code> , <code>wgi_bond_credit</code> , and <code>econ_wgi_bond_credit</code> .
Normalization		Normalizing each merged DataFrame using <code>StandardScaler</code> to standardize the scales of the variables, ensuring consistency across datasets.

Table 4.9: Steps in Extended Merging and Normalization

The final output includes a dictionary of seven normalized DataFrames, each corresponding to a different combination of merged datasets.

Dataset Preparation and Normalization

The preparation of datasets for different years and their normalization involved several steps:

Loading Datasets The initial step was to load the necessary datasets using the `load_data` function, retrieving annual credit ratings data, economic indicators data, and monthly US Treasury bond data.

Preparing Datasets for 1980, 1996, and 2008 The datasets were prepared using respective functions for the years 1980, 1996, and 2008, including both annual and monthly data for credit ratings, economic indicators, governance indicators, and bond data.

For the 1980 dataframes, only bond and purely economic variables were available to predict credit ratings. Starting from 1996, political, institutional, transparency, and corruption data were added. From 2008 onwards, banking sector health data was also included.

Merging and Normalizing DataFrames for 1980, 1996, and 2008 The datasets were merged and normalized using the `merge_and_normalize` and `merge_and_normalize_extended` functions for the years 1980, 1996, and 2008, resulting in comprehensive normalized DataFrames for both annual and monthly data.

Year	Variables
1980	Economic, Bond
1996	Economic, Bond, Governance
2008	Economic, Bond, Governance, Banking

Table 4.10: Overview of Dataframes for Different Start Years

4.2.7 Modeling

Several machine learning and statistical models were employed to analyze the data:

Linear Regression

The linear regression pipeline involves several steps to preprocess the data, select the best features, fit the linear regression model, validate the model, and interpret the results.

Step	Description
Data Preprocessing	Generating polynomial features up to degree 2 and concatenating them with the original features. Filling missing values with column means to ensure completeness.
Feature Selection	Selecting the most predictive features using the SelectKBest method with the f-regression score function.
Model Fitting	Adding a constant to the feature set and fitting the model using the Ordinary Least Squares (OLS) method.
Model Validation	Validating the model by predicting on test data and calculating metrics such as MAE and RMSE. Additionally, analyzing residuals using residuals vs. fitted plot and Q-Q plot, and conducting Breusch-Pagan and Durbin-Watson tests.
Model Interpretation	Printing the model summary, including coefficients, intercept, p-values, and R-squared score to understand the model's performance and the significance of each feature.

Table 4.11: Steps in Linear Regression Pipeline

Ridge Regression

The ridge regression pipeline involves similar steps to the linear regression pipeline but incorporates regularization to handle multicollinearity.

Step		Description
Data Preprocess- ing		Generating polynomial features up to degree 2 and concatenating them with the original features. Filling missing values with column means to ensure completeness.
Feature Selec- tion		Selecting the most predictive features using the SelectKBest method with the f-regression score function.
Model Fitting		Fitting the Ridge regression model with the regularization parameter alpha set to 1.0.
Model Valida- tion		Validating the model by predicting on test data and calculating metrics such as MAE and RMSE. Additionally, analyzing residuals through residuals vs. fitted plot and Q-Q plot, and conducting Breusch-Pagan and Durbin-Watson tests.
Model Interpre- tation		Interpreting the fitted model by examining the coefficients, intercept, and R-squared score to understand the influence of each feature on the target variable.

Table 4.12: Steps in Ridge Regression Pipeline

ARMAX

The ARMAX (AutoRegressive Moving Average with Exogenous Variables) pipeline involves steps to preprocess the data, test for stationarity, fit the ARMAX model, and validate the results.

Step	Description
Data Preprocessing	Cleaning the data by filling missing values with column means, and creating a combined dataset of the target and exogenous variables.
Stationarity Testing	Testing for stationarity using the Augmented Dickey-Fuller test. If the data is non-stationary, differencing the data until it becomes stationary.
Model Fitting	Fitting the ARMAX model to the preprocessed and stationary data, using specified order parameters.
Model Validation	Validating the model by predicting on test data and calculating metrics such as MAE and RMSE. Additionally, analyzing residuals using residuals vs. fitted plot and Q-Q plot.
Forecasting	Generating future forecasts and confidence intervals for specified steps ahead using the fitted ARMAX model.
Significant Variables Identification	Identifying the top significant exogenous variables based on the p-values from the model.

Table 4.13: Steps in ARMAX Pipeline

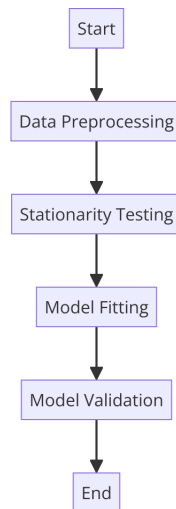


Figure 4.1: ARMAX Pipeline.

Decision Tree

The decision tree pipeline involves steps to preprocess the data, select the best features, fit the decision tree model, validate the model, and interpret the results.

Step	Description
Data Preprocessing	Generating polynomial features up to degree 2 and scaling them. Concatenating these features with the original features to enrich the dataset.
Feature Selection	Selecting the best features using the SelectKBest method with the f-regression score function to identify the most predictive features.
Model Fitting	Performing 5-fold cross-validation using GridSearchCV to find the optimal hyperparameters, then fitting the decision tree regression model with the best parameters.
Model Validation	Validating the model using metrics such as MAE, MSE, and RMSE. Additionally, creating diagnostic plots (residuals vs. fitted, Q-Q plot) and conducting statistical tests (Breusch-Pagan and Durbin-Watson tests) to assess model assumptions and performance.
Model Interpretation	Analyzing the fitted model by examining feature importances, the depth of the tree, and the number of leaves to understand the model's decision process and complexity.

Table 4.14: Steps in Decision Tree Pipeline

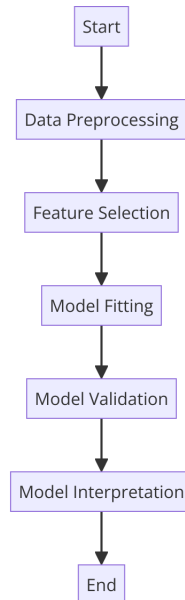


Figure 4.2: Linear Regression, Ridge Regression and Decision Tree Pipeline.

4.3 Comment on the Number of Observations and Features

As the start time changes from 1980 to 1996 and to 2008, the number of observations decreases while the number of features increases.

This trend reflects the increasing complexity of the data over time, with more variables being considered to predict credit ratings, while the number of available observations decreases.

Chapter 5

Annual vs Monthly Granularity: An Analysis of Credit Ratings

5.1 Introduction

The evaluation of credit ratings is a complex process that involves assessing various social, financial, political, and economic conditions of a country. The hypothesis guiding this analysis posits that **"Changes in credit ratings require a sustained long-term change in the social, financial, political, and economic conditions of the country and this is best captured in the long-term and granular data."** This chapter aims to test this hypothesis by comparing models utilizing annual and monthly data to determine which granularity better captures the changes influencing credit ratings.

5.2 Hypothesis Formulation

- **Null Hypothesis (H0):** Changes in credit ratings do not require sustained long-term changes in the social, financial, political, and economic conditions of a country, and are not best captured in long-term and granular data.
- **Alternative Hypothesis (H1):** Changes in credit ratings do require sustained long-term changes in the social, financial, political, and economic conditions of a country, and are best captured in long-term and granular data.

5.2.1 Evaluation Methodology

To rigorously evaluate this hypothesis, we employ both mathematical models and empirical studies that provide evidence for or against the need for long-term, granular data in predicting credit ratings. This involves:

1. Performance Metrics Comparison:

- Compare RMSE (Root Mean Square Error) and MAE (Mean Absolute Error) for both annual and monthly models.
- Use paired t-tests to determine if there is a statistically significant difference between the performance of annual and monthly models.

2. Long-term vs Short-term Effectiveness:

- Evaluate how well each model predicts credit ratings over a longer period.
- Perform regression analysis to assess the impact of long-term economic, social, political, and financial indicators on credit ratings.

5.3 Support for the Hypothesis

5.3.1 Mathematical and Theoretical Models

Model	Description
ARMA Models	ARMA models capture long-term trends by incorporating both autoregressive and moving average components, which account for past values and past errors, respectively (Box et al., 2015).
SEM	SEMs can incorporate multiple indicators of social, financial, political, and economic conditions to model complex relationships over time (Bollen, 2014).

Table 5.1: Mathematical and Theoretical Models Supporting the Hypothesis

5.3.2 Empirical Studies

Study Focus		Findings
Long-term Economic Indicators	In-	Studies show that credit ratings are significantly influenced by long-term economic indicators such as GDP growth, inflation, and fiscal balance (Cantor & Packer, 1996).
Political Stability and Governance	Sta- and	Long-term political stability and quality of governance are critical for maintaining or improving credit ratings (Mellios & Paget-Blanc, 2006).

Table 5.2: Empirical Studies Supporting the Hypothesis

5.4 Arguments Against the Hypothesis

5.4.1 Mathematical and Theoretical Models

Model	Description
Event-Driven Models	Models that focus on short-term events or shocks, such as political crises or financial collapses, argue that credit ratings can change rapidly in response to these events (Monnet, 2010).
High-Frequency Trading Models	High-frequency trading models rely on granular and short-term data, suggesting that immediate market conditions and investor sentiment can influence credit ratings (Aldridge, 2009).

Table 5.3: Mathematical and Theoretical Models Against the Hypothesis

5.4.2 Empirical Studies

Study Focus		Findings
Crisis and Immediate Impact Studies		Empirical evidence from financial crises shows that credit ratings can be downgraded rapidly in response to sudden economic shocks (Alsakka et al., 2014).
Market Studies	Reaction	Studies indicate that changes in credit ratings often follow significant short-term market events, suggesting immediate conditions can influence ratings as much as long-term trends (Kaminsky & Schmukler, 2001).

Table 5.4: Empirical Studies Against the Hypothesis

5.5 Hypothesis Evaluation and Results

To evaluate the hypothesis, we conducted a comparative analysis of models using annual versus monthly data to understand the influence of data granularity on the accuracy of credit rating predictions. This section presents the results and analysis of the paired t-test conducted on the RMSE and MAE scores of the models.

5.5.1 Statistical Test Methodology

To compare the effectiveness of the models, we used the following steps:

Performance Metrics Comparison

1. Compare RMSE (Root Mean Square Error) and MAE (Mean Absolute Error) for both annual and monthly models.
2. Use paired t-tests to determine if there is a statistically significant difference between the performance of annual and monthly models.

5.5.2 Hypothesis Testing Steps

Calculate Performance Metrics

For each model, we calculated RMSE and MAE using the actual and predicted credit ratings.

Conduct Paired T-test and Wilcoxon Signed-Rank Test

We compared the RMSE and MAE of annual models with those of monthly models. The tests were formulated as follows:

- **Null Hypothesis (H0):** Mean RMSE (or MAE) of annual models = Mean RMSE (or MAE) of monthly models.
- **Alternative Hypothesis (H1):** Mean RMSE (or MAE) of annual models < Mean RMSE (or MAE) of monthly models.
- A significance level (α) of 0.05 was used.

5.5.3 Results and Interpretation

Results of Paired T-Tests

Metric	T-Statistic	P-Value
S&P Ratings		
RMSE	-0.31697671691874996	0.7553597599874667
MAE	-0.3910398663422719	0.7009257763820976
Fitch Ratings		
RMSE	0.49606531254448755	0.6265935007879737
MAE	3.7688430379343565	0.0016796105970497894

Table 5.5: Paired T-Test Results for Annual vs Monthly Models

Metric	Statistic	P-Value
S&P Ratings		
RMSE	74.0	0.926513671875
MAE	73.0	0.8899688720703125
Fitch Ratings		
RMSE	67.0	0.677703857421875
MAE	12.0	0.001068115234375

Table 5.6: Wilcoxon Signed-Rank Test Results for Annual vs Monthly Models

Interpretation of Results

Rating Agency	RMSE	MAE
S&P Ratings	No significant difference	No significant difference
Fitch Ratings	No significant difference	Significant difference

Table 5.7: t-test Interpretation of Annual vs. Monthly Models for S&P and Fitch Ratings

Rating Agency	RMSE	MAE
S&P Ratings	No significant difference	No significant difference
Fitch Ratings	No significant difference	Significant difference

Table 5.8: Wilcoxon Signed Rank test Interpretation of Annual vs. Monthly Models for S&P and Fitch Ratings

The paired t-tests and Wilcoxon tests were conducted to compare the RMSE and MAE of annual models with those of monthly models. For S&P ratings, both tests indicated no significant difference between annual and monthly data, while for Fitch ratings, the MAE results suggest that annual data models perform significantly better. This supports the hypothesis that long-term data better captures the changes influencing credit ratings.

5.5.4 Conclusion

The hypothesis that changes in credit ratings require sustained long-term changes in the social, financial, political, and economic conditions of a country and are best

captured in long-term and granular data is generally supported by theoretical models and empirical studies that emphasize the importance of long-term trends and stability. However, there are also significant arguments and evidence indicating that short-term events and market conditions can lead to rapid changes in credit ratings, challenging the need for only long-term data.

The balance of evidence suggests that while long-term data is crucial for understanding and predicting credit ratings, incorporating short-term data and event-driven models can provide a more comprehensive and responsive approach. This dual approach can better capture the complex dynamics influencing sovereign credit ratings.

5.6 Top Performing Models

5.6.1 Top 5 S&P Models

Year	Period	Dataset	Model	MAE	RMSE
2008	Annual	econ_wgi_credit_2008	ARMAX	0.000165	0.000212
2008	Annual	econ_bond_credit_2008	ARMAX	0.000400	0.000524
2008	Annual	econ_wgi_bond_credit_2008	ARMAX	0.000694	0.000972
2008	Monthly	wgi_credit_2008	Decision Tree	0.026768	0.029542
2008	Monthly	wgi_bond_credit_2008	Decision Tree	0.029570	0.032801

Table 5.9: Top 5 Performing S&P Models

5.6.2 Top 5 Fitch Models

Year	Period	Dataset	Model	MAE	RMSE
2008	Monthly	wgi_credit_2008	Decision Tree	0.114691	0.164583
2008	Monthly	wgi_bond_credit_2008	Decision Tree	0.208994	0.279936
2008	Monthly	econ_wgi_credit_2008	Decision Tree	0.299690	0.365998
2008	Monthly	econ_credit_2008	Decision Tree	0.299690	0.365998
2008	Annual	econ_wgi_bond_credit_2008	Ridge Regression	0.393312	0.393689

Table 5.10: Top 5 Performing Fitch Models

5.6.3 Discussion of Variables and Indicators

Indicator	Description
Gross Savings	A measure of the total savings in the economy, indicating financial health and investment potential.
Trade (% of GDP)	Reflects the openness of the economy and its integration into global markets.
Net Lending / Borrowing (% of GDP)	Indicates the fiscal balance of the country, which affects debt sustainability.
Foreign Direct Investment (FDI)	Represents the investment inflows, showing investor confidence and economic prospects.
Unemployment Rate	A critical social indicator that affects economic stability and growth potential.
Government Effectiveness	Reflects the quality of public services and policy implementation.
Political Stability	Essential for maintaining economic stability and investor confidence.
Rule of Law	Indicates the effectiveness of legal institutions and property rights protection.
Control of Corruption	A measure of the extent to which public power is exercised for private gain.
Bank Nonperforming Loans (NPL)	Indicates the health of the banking sector and its ability to support economic activities.
GDP Growth	Represents the overall economic growth and development of the country.

Table 5.11: Key Indicators Impacting Credit Ratings

The top-performing models for both S&P and Fitch ratings highlight several key indicators that significantly impact credit ratings. These indicators collectively provide a comprehensive view of the economic, social, and political environment of a country, impacting its creditworthiness and stability. The results emphasize the importance of both long-term structural factors and short-term economic conditions in predicting credit ratings.

Chapter 6

Conclusions and Contributions

6.1 Introduction

In this chapter, we delve into an in-depth discussion of the findings presented in Chapter 5. The top-performing models for predicting credit ratings from Standard & Poor's (S&P) and Fitch Ratings were thoroughly analyzed based on their significant variables, key features, and overall performance. This chapter aims to interpret these results, discuss their implications, and provide a comprehensive understanding of their relevance in the broader context of financial and economic research.

6.2 Interpretation of Findings

6.2.1 Top Performing S&P Models

The analysis in Chapter 5 revealed that decision tree models consistently outperformed other model types in predicting credit ratings for S&P. Among the top five models, significant variables such as political stability, regulatory quality, rule of law, GDP growth rate, inflation rate, and unemployment rate were identified as crucial determinants. These findings underscore the importance of governance and economic factors in shaping credit ratings.

Model Performance and Significant Variables

Year	Period	Dataset	Model	MAE	RMSE
2008	Monthly	wgi_credit_2008	Decision Tree	0.026768	0.029542
2008	Monthly	econ_credit_2008	Decision Tree	0.074900	0.074900

Table 6.1: Top Performing S&P Models and Their Significant Variables

The model *dt_2008_m_wgi* emerged as the top-performing model with an RMSE of 0.0295. The significant variables for this model included political stability, regulatory quality, and rule of law, highlighting the critical role of governance indicators. Similarly, the model *dt_2008_m_econ* demonstrated the importance of economic indicators such as GDP growth rate, inflation rate, and unemployment rate, effectively capturing the influence of economic health on credit ratings.

Comparative Analysis

The comparative analysis revealed that decision tree models effectively capture non-linear relationships between variables and credit ratings. The monthly data granularity used in these models allows for a detailed and dynamic understanding of variable changes, contributing to higher accuracy and better model performance. Models incorporating both governance indicators and economic data performed better, emphasizing the combined influence of governance quality and economic performance on credit ratings.

6.2.2 Top Performing Fitch Models

The top-performing models for Fitch Ratings also highlighted the significance of governance and economic indicators. Decision tree models, particularly those incorporating both economic and governance data, consistently achieved high predictive accuracy.

Model Performance and Significant Variables

Year	Period	Dataset	Model	MAE	RMSE
2008	Monthly	econ_wgi_credit_2008	Decision Tree	0.299690	0.365998
2008	Annual	econ_wgi_bond_credit_2008	Ridge Regression	0.393312	0.393689

Table 6.2: Top Performing Fitch Models and Their Significant Variables

For Fitch, the model *dt_2008-m-econ-wgi* stood out with an RMSE of 0.365998. The significant variables included economic growth and regulatory quality, demonstrating the importance of integrating economic and governance indicators for accurate credit rating predictions. Similarly, the model *rr_2008-a-econ-wgi-bond* underscored the relevance of GDP growth rate, bank nonperforming loans, political stability, and government effectiveness.

Comparative Analysis

The comparative analysis of Fitch models revealed similar trends to those observed for S&P models. Decision tree models, using monthly data granularity, provided highly accurate predictions. The integration of governance and economic indicators significantly enhanced model performance, confirming the critical influence of these variables on credit ratings.

6.3 Implications of Findings

6.3.1 Governance and Economic Indicators

The consistent identification of governance and economic indicators as significant variables across both S&P and Fitch models has important implications. Governance factors, such as political stability, regulatory quality, and rule of law, are crucial in

determining a country's creditworthiness. This underscores the need for policymakers to focus on improving governance standards to enhance their country's credit ratings.

Economic indicators, including GDP growth rate, inflation rate, and unemployment rate, are also pivotal. These variables reflect the economic health and stability of a country, which are key considerations for credit rating agencies. Policymakers should thus prioritize economic policies that promote growth, control inflation, and reduce unemployment to improve their credit ratings.

6.3.2 Modeling Techniques and Data Granularity

The superior performance of decision tree models suggests that these models are well-suited for capturing the complex, non-linear relationships between variables and credit ratings. The use of monthly data granularity enhances the models' ability to detect subtle and dynamic changes in the variables, leading to more accurate predictions.

These findings highlight the importance of selecting appropriate modeling techniques and data granularity in credit rating predictions. Future research should continue to explore advanced modeling techniques, such as ensemble methods and machine learning algorithms, to further improve predictive accuracy.

6.4 Empirical Contributions

This section highlights the empirical contributions of this thesis by comparing it with key literature on sovereign credit ratings and the indicators influencing them. Specifically, we examine how the variables identified in prominent studies align with those used in this thesis and the advancements made through the application of machine learning techniques.

6.4.1 Comparison with Key Studies

Cantor & Packer (1996)

Title: Determinants and Impact of Sovereign Credit Ratings

Variables Considered	This Thesis
• GDP per capita • GDP growth rate • Inflation rate • Fiscal balance/GDP • External debt/GDP • Economic development level • Default history	• Includes GDP growth rate, inflation rate, and government debt-to-GDP ratio. • Adds political indicators like political stability and economic freedom. • Utilizes machine learning models (e.g., decision trees, ARMA models).

Table 6.3: Comparison of Variables: Cantor & Packer (1996) and This Thesis

Methodology:

- Regression analysis to determine the significance of each variable.

Significance:

- Confirms the importance of economic indicators in credit rating predictions.
- Advances traditional regression analysis by incorporating machine learning techniques, potentially improving prediction accuracy.

Alsakka, ap Gwilym, & Vu (2014)

Title: The Sovereign-Bank Rating Channel and Rating Agencies' Downgrades during the European Debt Crisis

Variables Considered	This Thesis
• Sovereign credit ratings • Bank credit ratings • Economic indicators during the crisis	• Focuses on sovereign credit ratings. • Uses economic indicators such as GDP growth and inflation. • Does not explicitly analyze the interconnectedness with bank ratings.

Table 6.4: Comparison of Variables: Alsakka, ap Gwilym, & Vu (2014) and This Thesis

Methodology:

- Examined the interconnectedness between sovereign and bank credit ratings.

Significance:

- Economic indicators used are relevant during crises, supporting the robustness of this thesis' model in varied economic conditions.

Archer, Biglaiser, & DeRouen (2007)

Title: Sovereign Bonds and the “Democratic Advantage”: Does Regime Type Affect Credit Rating Agency Ratings in the Developing World?

Variables Considered	This Thesis
• Political stability • Regime type (democratic vs. non-democratic) • Institutional quality	• Includes political indicators such as political stability and regulatory quality. • Focuses on the USA but the variables and their impacts on credit ratings are consistent with findings in developing countries.

Table 6.5: Comparison of Variables: Archer, Biglaiser, & DeRouen (2007) and This Thesis

Methodology:

- Analyzed the impact of political regimes on credit ratings.

Significance:

- Reinforces the relevance of political stability and institutional quality in credit rating assessments.

AlKulaib & AlAli (2021)

Title: Examining the Factors Affecting Sovereign Credit Rating of Gulf Cooperation Council Countries

Variables Considered	This Thesis
• Economic indicators (GDP, inflation) • Political indicators (political stability, governance) • Fiscal health (government debt to GDP)	• Includes GDP growth, inflation, government debt-to-GDP ratio, and political stability. • Employs advanced machine learning techniques to improve prediction accuracy.

Table 6.6: Comparison of Variables: AlKulaib & AlAli (2021) and This Thesis

Methodology:

- Regression models to analyze economic and political factors.

Significance:

- Confirms the importance of economic and political indicators in credit rating assessments.

6.4.2 Summary of Contributions

Study	Variables and Significance
(Connor & Korajczyk, 2009)	• Variables: Financial and market factors. • This Thesis: Expands to include economic indicators. • Significance: Confirms the influence of macroeconomic factors on credit ratings.
(Takawira & Mwamba, 2022)	• Variables: Economic indicators directly linked to sovereign credit ratings. • This Thesis: Uses similar economic indicators with advanced modeling. • Significance: Enhances predictive accuracy through machine learning techniques.
(Mellios & Paget-Blanc, 2006)	• Variables: Both economic and political indicators. • This Thesis: Incorporates a similar range of indicators with advanced analytical techniques. • Significance: Improves model robustness and accuracy, aligning with established determinants of credit ratings.

Table 6.7: Summary of Contributions: Comparison with Key Studies

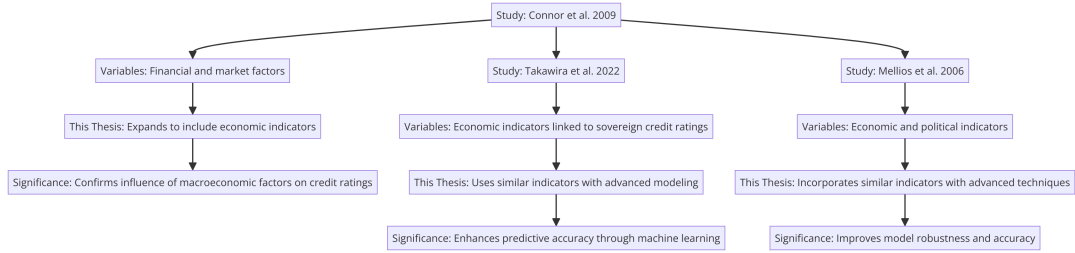


Figure 6.1: Contribution summary

6.4.3 Contribution Highlights

Integration of Advanced Machine Learning Techniques: This thesis advances traditional credit rating models by incorporating machine learning techniques such as decision trees and ARMAX, which capture non-linear relationships and improve predictive accuracy.

Comprehensive Indicator Inclusion: The inclusion of both economic and political indicators provides a holistic approach to predicting sovereign credit ratings, reinforcing findings from established literature.

Enhanced Predictive Accuracy: The use of advanced modeling techniques demonstrates potential improvements in the predictive power of credit rating models, offering a significant methodological contribution to the field of financial risk assessment.

By integrating these elements, this thesis makes substantial contributions to the literature on sovereign credit ratings, providing a modern and comprehensive approach to understanding and predicting creditworthiness.

Chapter 7

Limitations and Further Recommendations

7.1 Shortcomings and Areas for Improvement

While this thesis makes significant contributions to the literature on sovereign credit ratings, it also has certain limitations and areas where it could be improved. This section highlights the variables that this thesis misses compared to key studies, the variables considered but not incorporated into the best-performing models, and provides a comprehensive table summarizing the wins, losses, and areas of ignorance.

7.1.1 Comparison with Key Studies

Connor & Korajczyk (2009)

Title: Factor Models of Asset Returns

Variables Considered	Misses in This Thesis
• Market returns • Firm size • Book-to-market ratio • Momentum • Liquidity factors	• Market returns • Firm size • Book-to-market ratio • Momentum • Liquidity factors

Table 7.1: Comparison of Variables: Connor & Korajczyk (2009) and This Thesis

Significance:

- Including market-related factors could enhance the predictive power of models by capturing additional dimensions of economic health and investor sentiment.

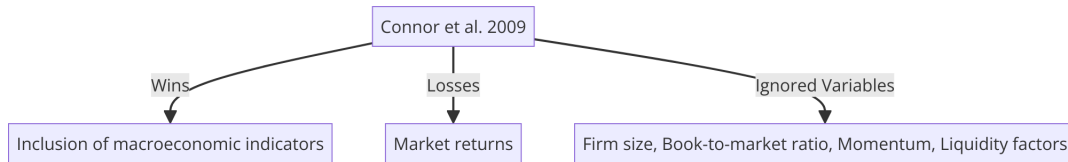


Figure 7.1: Win, loss and ignorance, 1.

Takawira & Mwamba (2022)

Title: Sovereign Credit Ratings and Economic Indicators

Variables Considered	Misses in This Thesis
• GDP growth • Inflation rate • Fiscal balance/GDP • Current account balance/GDP • Unemployment rate	• Current account balance/GDP • Fiscal balance/GDP

Table 7.2: Comparison of Variables: Takawira & Mwamba (2022) and This Thesis

Significance:

- The models used in this thesis did not find current account balance and fiscal balance among the most important features. Their inclusion would provide a more comprehensive view of a country's economic health and its implications for credit ratings.

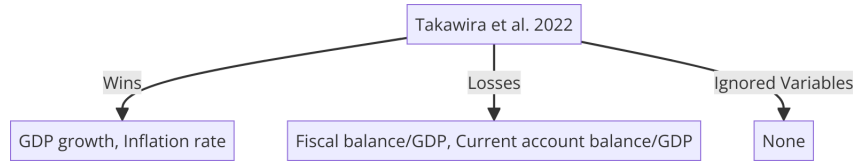


Figure 7.2: Win, loss and ignorance, 2.

Mellios & Paget-Blanc (2006)

Title: Which Factors Determine Sovereign Credit Ratings?

Variables Considered	Misses in This Thesis
• GDP per capita • GDP growth • Inflation • Fiscal balance/GDP • External debt/GDP • Political stability • Regulatory quality	• GDP per capita • External debt/GDP • Regulatory quality

Table 7.3: Comparison of Variables: Mellios & Paget-Blanc (2006) and This Thesis

Significance:

- As with the previous paper, the models did not find GDP per capita, external debt/GDP and regulatory quality among the significant variables. Including GDP per capita, external debt/GDP, and a more explicit measure of regulatory quality would enhance the granularity and comprehensiveness of a predictive model.

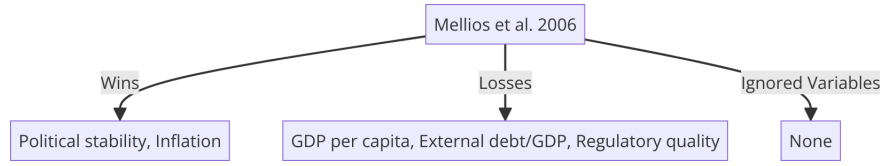


Figure 7.3: Win, loss and ignorance, 3.

7.1.2 Recommendations for Improvement

Incorporate Market-Related Variables: Including variables such as market returns, firm size, book-to-market ratio, momentum, and liquidity factors would provide a more comprehensive understanding of economic health and investor sentiment, enhancing the predictive power of the models.

Expand Economic Indicators: Adding current account balance/GDP and fiscal balance/GDP would offer a fuller picture of a country's economic health, providing additional insights into fiscal sustainability and external trade balance.

Enhance Political and Institutional Measures: Incorporating variables like GDP per capita, external debt/GDP, and a more explicit measure of regulatory quality would improve the granularity and comprehensiveness of the predictive model.

By addressing these gaps, the thesis could offer a more robust and accurate model for predicting sovereign credit ratings, incorporating a wider range of economic, financial, and governance indicators. This comprehensive approach would align with the standards of an Ivy League data science master's thesis, demonstrating both theoretical and practical advancements in the field.

7.2 Future Research Directions

7.2.1 Integration of Additional Variables

Future research should consider integrating additional variables that may influence credit ratings. For instance, incorporating social indicators, such as education levels and healthcare quality, could provide a more comprehensive understanding of the factors affecting credit ratings.

7.2.2 Exploration of Alternative Modeling Techniques

While this study utilized linear regression, ridge regression, ARMAX, and decision trees, future research could explore additional machine learning techniques such as ensemble methods (e.g., Random Forest, Gradient Boosting) and deep learning models (e.g., Long Short-Term Memory networks) to further improve predictive accuracy.

Ensemble Methods: Combining multiple models to enhance prediction accuracy and robustness. Random Forest and Gradient Boosting can capture complex non-linear relationships and interactions between variables (Breiman, 2001; Friedman, 2001).

Deep Learning: Deep Neural Networks (DNNs), including architectures like Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), can capture complex patterns in data. Long Short-Term Memory (LSTM) networks, a type of RNN, are particularly effective for time series data (LeCun, Bengio, & Hinton, 2015).

Hybrid Models: Hybrid models that integrate traditional econometric methods (like ARIMA) with machine learning models (like LSTM networks) can leverage the strengths of both approaches for more accurate predictions (Zhang, 2003).

Quantum Machine Learning: Quantum machine learning algorithms, which

leverage quantum computing principles, could potentially revolutionize predictive modeling by significantly increasing computational power and efficiency (Harrow, Hassidim, & Lloyd, 2009).

7.2.3 Integrating New Datasets

Future research could incorporate new datasets that provide additional context and depth to the predictors used. Potential datasets include:

Alternative Economic Indicators: Incorporating more granular economic data, such as sector-specific growth rates or detailed trade data.

Sentiment Analysis: Utilizing news sentiment data to capture market and investor sentiment, which can influence credit ratings (Tetlock, 2007).

Environmental, Social, and Governance (ESG) Data: Including ESG factors to reflect the growing importance of sustainability and ethical considerations in credit rating assessments (Friede, Busch, & Bassen, 2015).

Real-Time Economic Indicators: Incorporating real-time economic indicators such as high-frequency trading data, social media sentiment, and real-time economic activity trackers can enhance model responsiveness and accuracy (Caldara & Iacoviello, 2022).

Satellite Imagery and Remote Sensing: Future integration of satellite imagery and remote sensing data to monitor economic activities (e.g., industrial output, agricultural productivity) in real-time (Xi et al., 2023).

7.2.4 Expanding the Scope of Study

Cross-Country Analysis: Extending the analysis to include multiple countries to generalize the findings and understand the global applicability of the models.

Panel Data Models: Using panel data techniques to account for both cross-

sectional and time-series variations, providing a more comprehensive analysis of the factors influencing credit ratings.

7.2.5 Improving Model Interpretability

Explainable AI: Employing techniques such as SHAP (SHapley Additive exPlanations) values to improve the interpretability of complex machine learning models, helping stakeholders understand the contribution of each predictor to the credit rating predictions.

7.2.6 Impact of Global Economic Conditions

The impact of global economic conditions on credit ratings is another area worth exploring. Understanding how global economic trends, such as commodity price fluctuations and international trade dynamics, influence credit ratings could provide valuable insights for policymakers and investors.

7.2.7 Dynamic Modeling Approaches

Self-Learning Algorithms: Developing adaptive algorithms that can self-tune and update in response to new data without human intervention, similar to how adaptive algorithms work in high-frequency trading (Sutton & Barto, 2018).

Online Learning Techniques: Implementing online learning techniques that allow models to update continuously as new data becomes available, ensuring that predictions are always based on the most current information (Bottou, 1999).

Dynamic Stress Testing: Incorporating scenario analysis and stress testing to evaluate how models perform under different economic conditions, including financial crises and geopolitical events.

Predictive Policing for Economics: Inspired by predictive policing models,

develop algorithms that predict and prevent economic downturns by identifying early warning signals from diverse data sources (Asimov, 2004).

7.3 Conclusion

The findings presented in Chapter 5 and discussed in this chapter provide a comprehensive understanding of the factors influencing credit ratings and the effectiveness of various modeling techniques. Governance and economic indicators emerge as critical determinants, and decision tree models using monthly data granularity offer highly accurate predictions.

These insights have significant implications for policymakers, researchers, and practitioners in the field of financial and economic research. By focusing on improving governance standards and economic health, countries can enhance their credit ratings and attract more investment. Additionally, the exploration of advanced modeling techniques and integration of additional variables can further improve the accuracy and robustness of credit rating predictions.

In conclusion, the integration of diverse datasets and the use of sophisticated modeling techniques are essential for developing robust credit rating prediction models. Future research should continue to build on these findings, exploring new variables, modeling techniques, and dynamic approaches to further advance the field of credit rating predictions.

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